SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Purpose and Scope

This Software Requirements Specification (SRS) defines the complete requirements for TrackWIL version 1.0, the system built and tested in this project. Chapter 3 summarises these requirements, Chapter 4 shows how they were designed and implemented, and Chapter 5 traces every requirement to its test cases in the Requirements Traceability Matrix.

TrackWIL is a web-based system that replaces the paper, PDF and email process for Work-Integrated Learning (WIL) logbooks at CTU Training Solutions. Students record daily work and evidence, workplace mentors verify each entry through a one-time link without needing an account, and coordinators and lecturers monitor every student's progress on a shared dashboard.

In scope (version 1.0): student sign-up and sign-in; WIL placements with their own required hours; daily logbook entries with optional evidence; WIL documents for each placement (MOU, Commencement Letter, Code of Conduct and an attendance register per semester); submission for verification through a one-time link; mentor approval or rejection; entry statuses and verified-hour totals; the coordinator and lecturer dashboard with a profile for each student; WIL completion; an SQLite database.

Out of scope (version 1.0): mentor accounts, written mentor feedback, coordinator review of uploaded documents, notifications, editing or deleting entries, reports and exports, integration with CTU systems, a mobile application and the use of real personal data.

Table 1: Definitions

Term	Meaning
WIL	Work-Integrated Learning: the workplace component of a qualification
Entry	One day of WIL work recorded by a student for one placement (date, description, hours)
Verification link / token	A one-time URL containing a 48-character random token that lets a mentor verify one entry without signing in
Verified hours	The total hours of a placement's entries with the status Verified
Overdue	A student whose current placement has no verified entry in the last 14 days (counted from the placement start if nothing is verified yet) and no entry awaiting verification
WIL placement	One period of WIL at a host company, with its own required hours, logbook and documents
WIL documents	The Memorandum of Understanding (MOU), Commencement Letter and Code of Conduct (once per placement) and the Attendance Register (once per semester)
WIL completed	A placement whose verified hours have reached its required hours; it is then locked

2. Users and Stakeholders

Table 2: User classes

User class	Description	Access	Main tasks
Student	Enrolled CTU student completing WIL hours	Account (student)	Sign up, add WIL placements, record entries, upload evidence and WIL documents, request verification, track hours
Workplace mentor	Staff member at the host organisation who supervises the student	One-time link only	Review and approve or reject an entry
Coordinator	CTU staff member who manages WIL compliance	Account (coordinator)	Monitor all students' status, hours and documents; open student profiles
Lecturer	CTU lecturer who supports students academically	Account (lecturer)	Monitor all students' status, hours and documents; open student profiles

Secondary stakeholders are the training institution (needs reliable records and POPIA compliance), the development team (needs clear, testable requirements) and the project supervisor (checks alignment with the project objectives and rubric).

3. Functional Requirements

Table 3: Functional requirements with acceptance criteria

ID	Requirement	Priority	Acceptance criterion
FR-01	Sign in with email and password and go to the dashboard for the user's role.	High	Correct details open /student for students and /coordinator for coordinators and lecturers; wrong details show "Invalid email or password." and no session is created. Email matching ignores letter case.
FR-02	Student sign-up with a CTU student email (eight digits @ctucareer.co.za) and a strong password.	Medium	Password needs 8+ characters with upper case, lower case, a number and a special character; a non-CTU or duplicate email is rejected with a message; a valid sign-up signs the student in.
FR-03	Role-based access to pages and actions.	High	Signed-out visitors are sent to the sign-in page; coordinators and lecturers cannot open the student area; students cannot open the dashboard; a student cannot view another student's entry or placement; a blocked page sends none of its data.

ID	Requirement	Priority	Acceptance criterion
FR-04	Create a daily logbook entry.	High	The entry belongs to one of the student's open placements; work date, description and hours are required; hours must be more than 0 and at most 24; the entry is saved as Draft and opened.
FR-05	Attach one optional evidence file to an entry.	Medium	PDF, PNG, JPG, JPEG and WEBP files are accepted and stored under a random name; any other type is rejected with "Evidence must be a PDF or image file."
FR-06	Submit an entry for verification.	High	Only Draft or Rejected entries with a mentor email can be submitted; a 48-character token valid for 7 days is created; the entry becomes Pending verification and the link is shown with a Copy button.
FR-07	Mentor verification through the link without an account.	High	The link shows the student's name, work date, hours, description and status, with Approve and Reject buttons; Approve sets Verified and records the time; Reject sets Rejected.
FR-08	One-time, expiring links.	High	A used link shows "This link was already used"; an expired or unknown link shows an error; neither changes the entry; two clicks on Approve record only one decision.
FR-09	Show entry status.	High	Each entry shows Draft, Pending verification, Verified or Rejected on the logbook and entry pages.
FR-10	Logbook history and hour progress.	High	Each placement's logbook lists its entries newest first and shows "X of Y verified hours completed" against the placement's required hours.
FR-11	Coordinator and lecturer monitoring dashboard.	High	Every student is listed with their status for their current placement (On track, Awaiting verification, Overdue, WIL completed or No placement), verified hours against required hours, number of entries, number pending and missing documents.
FR-12	Automatic timestamps.	Medium	created_at, updated_at and verified_at are stored on entries; used_at and decision are stored on links.
FR-13	Protected evidence download.	High	The student who uploaded a file, coordinators and lecturers can open it; another student receives 403 Forbidden and a signed-out visitor receives 401 Unauthorized.

ID	Requirement	Priority	Acceptance criterion
FR-14	Sign out.	High	Sign out removes the session cookie and returns to the sign-in page; protected pages then redirect to sign-in.
FR-15	WIL placements with their own required hours.	High	A student adds a placement with a title, company, start date and required hours (more than 0 and at most 5 000); invalid details are rejected with a message; a placement can be edited until it is completed; entries can only be added to the student's own placements, and another student's placement cannot be opened.
FR-16	WIL documents for each placement.	High	The MOU, Commencement Letter and Code of Conduct are kept once per placement and the Attendance Register once per semester; PDF, PNG and JPG files up to 10 MB whose content matches their type are accepted; uploading again replaces the file; the student can remove a document; only the student, coordinators and lecturers can open it.
FR-17	Student profile for coordinators and lecturers.	High	Selecting a student on the dashboard opens their profile with each placement's verified hours, documents (view only) and logbook; students cannot open it; an unknown student or a staff account returns 404 Not Found.
FR-18	WIL completion.	High	A placement is marked WIL completed as soon as its verified hours reach its required hours; new entries, submissions, document changes and edits on it are refused with a message; the student is asked to add the next placement.

4. Non-Functional Requirements

Table 4: Non-functional requirements

ID	Category	Requirement	Measure
NFR-01	Usability	A first-time student can create and submit an entry for verification in under 3 minutes without help.	Timed task (Chapter 5)
NFR-02	Performance	Pages load and actions complete within 2 seconds on the test computer with the demonstration data.	Browser timing

ID	Category	Requirement	Measure
NFR-03	Security	Salted scrypt password hashes; HttpOnly session cookie; tokens with at least 128 bits of randomness; server-side role checks on every protected page and action.	Code review and security tests
NFR-04	Reliability and integrity	Database constraints reject invalid data; multi-step updates are atomic transactions.	Constraint tests
NFR-05	Maintainability	TypeScript, layered modules, lint passes, README setup in five commands or fewer.	Lint and README walkthrough
NFR-06	Compatibility	Works in current Chrome and Edge; usable at 375 px and 1280 px wide.	Browser and responsive tests
NFR-07	Privacy	Only data needed for WIL tracking is stored; synthetic data only until written consent.	Data review

5. Use Cases / User Stories

Table 5: Use cases

ID	Use case	Actor(s)	Requirements
UC-01	Sign in / sign up	Student, coordinator, lecturer	FR-01, FR-02, FR-03
UC-02	Create logbook entry	Student	FR-04, FR-09, FR-12
UC-03	Upload evidence (extends UC-02)	Student	FR-05, FR-13
UC-04	Submit for verification	Student	FR-06
UC-05	Approve or reject via link	Workplace mentor	FR-07, FR-08
UC-06	View monitoring dashboard	Coordinator, lecturer	FR-10, FR-11
UC-07	Sign out	Student, coordinator, lecturer	FR-14
UC-08	Manage WIL placements	Student	FR-15, FR-18
UC-09	Upload WIL documents	Student	FR-16
UC-10	View student profile (extends UC-06)	Coordinator, lecturer	FR-03, FR-17

The user stories US-01 to US-11 and their acceptance criteria are listed in Chapter 3, Section 3.7.

6. Data Requirements

TrackWIL stores its data in an SQLite database with five tables. The full data dictionary is in the Database Design document.

- **users:** id, full name, email (unique, case-insensitive), role (student, coordinator or lecturer) and password hash.
- wil_placements: id, student id (foreign key to users), title, company name, required hours (more than 0 and at most 5 000), start date and the created and updated timestamps.

- **logbook_entries: id, placement id (foreign key to wil_placements),** work date, description, hours (more than 0 and at most 24), evidence file name, status, mentor email and the created, updated and verified timestamps.
- **verification_tokens:** token (primary key), entry id (foreign key to logbook_entries), expiry time, time used and decision (approved or rejected).

placement_documents: id, placement id (foreign key to wil_placements), document type (mou, commencement_letter, code_of_conduct or attendance_register), semester (attendance registers only), original file name, stored file name, file size (at most 10 MB) and upload time; one document of each type per placement and one attendance register per semester.

Validation rules: every rule is checked in the server action and, where possible, again by a database constraint (foreign key, UNIQUE or CHECK), so that invalid data cannot be stored even if a check in the code is missed.

Evidence files are stored in the data/uploads folder and WIL documents in data/uploads/documents, under random names; only the file names (and each document's original name and size) are stored in the database.

Retention: demonstration data is synthetic and can be reset at any time with npm run db:reset. If real data is used after permission is granted, personal information will be deleted when the project result is finalised, unless the campus sets another period.

7. Interfaces

Table 6: User interface screens

Screen	Route	Users
Sign in	/	All account holders
Sign up	/signup	New students
My WIL placements	/student	Students
Add or edit a placement	/student/placements/new, /student/placements/[id]/edit	Students
Placement logbook and documents	/student/placements/[id]	Students
New entry	/student/entries/new	Students
Entry details and verification link	/student/entries/[id]	Students
Mentor verification	/verify/[token]	Workplace mentors (no sign-in)
Coordinator dashboard	/coordinator	Coordinators and lecturers
Student profile	/coordinator/students/[id]	Coordinators and lecturers

Software interfaces: the application uses Node.js's built-in node:sqlite module to access the SQLite database file data/trackwil.db, and the server file system for evidence files and WIL documents. No external services are called.

Communication interfaces: browsers communicate with the server over HTTP (HTTPS in a hosted deployment). Forms call Next.js Server Actions; evidence files are served by the route /api/evidence/[fileName] and WIL documents by /api/documents/[id].

8. Constraints and Assumptions

- **Time:** the system had to be built, tested and demonstrated by 29 October 2026, so the scope was limited to the core workflow.
- **Technology:** Next.js 16 with TypeScript and Tailwind CSS; SQLite through Node.js 22.13 or newer; Git and GitHub for version control.
- **Legal and privacy:** POPIA applies; no real personal information is collected before written campus permission, so all testing uses synthetic data.
- **Deployment:** the prototype runs locally; cloud hosting is out of scope.
- **Assumption:** users have a modern web browser and an internet or network connection to the server.
- **Assumption:** students send the verification link to the correct mentor, and mentors check the work before approving it.
- **Assumption: students enter the correct required hours for each WIL placement (the demonstration placements require 120 or 200 hours).**

9. Acceptance Criteria

The solution is accepted when:

1. Every High-priority functional requirement (FR-01, FR-03, FR-04, FR-06 to FR-11 and FR-13 to FR-18) passes its test cases in the Test Cases and Results workbook.

2. A student can complete the full journey (sign in, open a placement, upload a document, create an entry with evidence, submit it, mentor approves through the link, hours increase) during the live demonstration using synthetic data.

3. Coordinators and lecturers can identify students who are on track, awaiting verification, overdue or WIL completed from the dashboard, and open each student's profile.

4. Security tests confirm that role restrictions, one-time links and protected evidence downloads work, and any open security defects are documented with a planned fix.

5. All defects found during testing are logged in the Defect Log and either fixed and retested or documented as known limitations.